

Course syllabus for MSE 201:002, Spring 2023

Course Information:

Course title: Structure and Properties of Engineering Materials

Course location: 01007, Engineering Building I

Meeting time: Tuesday/Thursday 1:30 PM-2:45 PM

Recorded lectures: **Panopto**

Class Slack: **Slack**

Instructor Information:

Name: Kaveh Ahadi

Office Location: 322, RB1

Office hours: Tuesday/Thursday 2:45 PM-3:30 PM or by appointment

Email: kahadi@ncsu.edu

TA Information:

Name: Mona Shabani

Office hours: Wednesday 3:30 PM-4:30 PM

Email: mshaban@ncsu.edu

Course overview:

Fundamentals of structure, structure modification and properties of materials with emphasis on structure-property relationships and modern theory of solids.

Learning outcomes:

Students completing this course should be able to:

Discuss, understand, and compare the structure of crystalline and amorphous solids

Identify the structure and chemistry of point, line, and planar defects in solids

Read and interpret a unary and binary phase diagram

Understand the crystalline lattice and the nomenclature used to describe it

Articulate the physical properties associated with ceramic, semiconducting, metallic, and amorphous solids

Understand the roles of thermodynamics and kinetics in regulating phase transitions

Textbook:

W. D. Callister and D.G. Rethwisch, Materials Science and Engineering: An Introduction, 8th Edition

Prerequisite:

Grade of C or better in CH 101.

Grading:

Test 1: 20% and Test 2: 20%

HW assignments: 20%

Final Exam (comprehensive): 40%

Tentative Schedule:

January 10th (Tu): Introduction to Materials Science (Chapter 1)

January 12th (Th): Atomic Structure and Interatomic Bonding (Chapter 2)

January 17th (Tu): Crystallography and Diffraction (Chapter 3)

January 19th (Th): Crystallography and Diffraction (Chapter 3)/*HW 1 (Ch 2 and 3)*-EMA Conference (**Zoom**)

January 24th (Tu): Crystallography and Diffraction (Chapter 3)

January 26th (Th): Imperfections in Solids (Chapter 4)/**HW 1 due**/HW 2 (Ch 4)

January 31st (Tu): Imperfections in Solids (Chapter 4)

February 2nd (Th): Diffusion (Chapter 5)/**HW 2 due**/HW 3 (Ch 5)

February 7th (Tu): Diffusion (Chapter 5)

February 9th (Th): **Test 1**, chapters 1-5 (online/moodle)

February 14th (Tu): Mechanical Properties of Solids (Chapter 6)/**HW 3 due**

February 16th (Th): *NO CLASS* (Wellness Day)

February 21st (Tu): Mechanical Properties of Solids (Chapter 6)/HW 4 (Ch 6)

February 23rd (Th): Phase Diagrams (Chapter 9)

February 28st (Tu): Phase Diagrams (Chapter 9)/**HW 4 due**/HW 5 (Ch 9)

March 2nd (Th): Phase Diagrams (Chapter 9)/ Phase Transformations in Solids (Chapter 10)

March 7th (Tu): Phase Transformations in Solids (Chapter 10) /**HW 5 due**-APS Meeting (**Zoom**)

March 9th (Th): Phase Transformations in Solids (Chapter 10) /HW 6 (Ch 10)-APS Meeting (**Zoom**)

March 15th (Tu): *NO CLASS* (Spring Break)

March 16th (Th): *NO CLASS* (Spring Break)

March 21st (Tu): **Test 2**, chapters 6,9, and 10 (online/moodle)

March 23rd (Th): Applications and Processing of Materials (Chapter 11) /**HW 6 due**

March 28th (Tu): Introduction to Ceramics (Chapter 12)

March 30th (Th): Introduction to Polymers (Chapter 14)

April 4th (Tu): Introduction to Composites (Chapter 16)

April 6th (Th): Introduction to Electronic structure of Materials (Chapter 18) /HW 7 (Ch 18)

April 11th (Tu): Electrical Properties of Materials (Chapter 18)-MRS Spring (**Zoom**)

April 13th (Th): Magnetic Properties of Materials (Chapter 20)/**HW 7 due**-MRS Spring (**Zoom**)

April 18th (Tu): Optical Properties of Materials (Chapter 21)

April 20th (Th): *Final Exam review*

April 27th (Th): **Final Exam, 12-2:30pm, 01007, EB I** (comprehensive)

Grading scale:

The following grading scale will be used to determine the minimum grade for this course:

100	A+	88.0 – 89.9	B+	78.0 – 79.9	C+	68.0 – 69.9	D+	0 – 59.9	F
95.0 – 99.9	A	82.0 – 87.9	B	72.0 – 77.9	C	62.0 – 67.9	D		
90.0 – 95	A-	80.0 – 81.9	B-	70.0 – 71.9	C-	60.0 – 61.9	D-		

Requirements for credit-only grading (S/U):

To receive a grade of S, students are required to complete all exams and earn a grade of C- or better. Conversion from letter grading to credit only (S/U) grading is subject to university deadlines. Refer to the Registration and Records calendar for deadlines related to grading. Check the links below for enhanced S/U policy due to COVID-19:

<https://studentservices.ncsu.edu/your-resources/covid-19/spring2020-sat-grading/>

<https://studentservices.ncsu.edu/your-resources/covid-19/spring-2021/>

Requirements for Auditors (AU):

Information about and requirements for auditing a course can be found at

<http://policies.ncsu.edu/regulation/reg-02-20-04>.

Incomplete grades (IN):

Incomplete grades will be given only under extenuating circumstances, in accordance with NCSU policy as described at the website below. If an extended deadline is not authorized by the instructor or department, an unfinished incomplete grade will automatically change to an F after either (a) the end of the next regular semester in which the student is enrolled (not including summer sessions), or (b) by the end of 12 months if the student is not enrolled, whichever is shorter. Incompletes that change to F will count as an attempted course on transcripts. The burden of fulfilling an incomplete grade is the responsibility of the student.

http://www.ncsu.edu/policies/academic_affairs/grades_undergrad/REG02.50.3.php

Attendance:

Students are expected to view all lectures and to take all exams on the scheduled dates.

Academic integrity:

As a student enrolled in this class, you are expected to honor the NCSU policies regarding academic integrity found in the Code of Student Conduct at

http://www.ncsu.edu/policies/student_services/student_discipline/POL11.35.1.php

Students with disabilities:

Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with Disability Services Office (<http://www.ncsu.edu/dso/>) located at 1900 Student Health Center, Campus Box 7509, 515-7653. For more information on NC State's policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation at

http://www.ncsu.edu/policies/academic_affairs/courses_undergrad/REG02.20.1.php

Anti-discrimination statement:

NC State University provides equality of opportunity in education and employment for all students and employees. Accordingly, NC State affirms its commitment to maintain a work environment for all employees and an academic environment for all students that is free from all forms of discrimination. Discrimination based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation is a violation of state and federal law and/or NC State University policy and will not be tolerated. Harassment of any person (either in the form of quid pro quo or creation of a hostile environment) based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation also is a violation of state and federal law and/or NC State University policy and will not be tolerated. Retaliation against any person who complains about discrimination is also prohibited. NC State's policies and regulations covering discrimination, harassment, and retaliation may be accessed at

http://www.ncsu.edu/policies/campus_environ

or

http://www.ncsu.edu/equal_op

Any person who feels that he or she has been the subject of prohibited discrimination, harassment, or retaliation should contact the Office for Equal Opportunity (OEO) at 515-3148.

Web resources:

A wolflware moodle site will be maintained for this course.

Online course evaluations:

Online class evaluations will be available for students to complete during the last two weeks of class. All evaluations are confidential; instructors will never know how any one student responded to any question, and students will never know the ratings for any instructor. The class evaluation website is

<https://classeval.ncsu.edu>

More information regarding the online class evaluation process is available at

<http://www2.acs.ncsu.edu/UPA/classeval/index.htm>

Students are required to comply with the university policy on academic integrity found in the Code of Student Conduct found at <http://policies.ncsu.edu/policy/pol-11-35-01>. Cheating and plagiarism will result in loss of credit for the assignment in question and filing of the Report of an Academic Integrity Violation. Students are responsible for reviewing the PRRs which pertain to their course rights and responsibilities. These include: <http://policies.ncsu.edu/policy/pol-04-25-05> (Equal Opportunity and Non-Discrimination Policy Statement), <http://oied.ncsu.edu/oied/policies.php> (Office for Institutional Equity and Diversity), <http://policies.ncsu.edu/policy/pol-11-35-01> (Code of Student Conduct), and <http://policies.ncsu.edu/regulation/reg-02-50-03> (Grades and Grade Point Average).

Electronically-Hosted Course Components. Students may be required to disclose personally identifiable information to other students in the course, via electronic tools like email or web-postings, where relevant to the course. Examples include online discussions of class topics, and posting of student coursework. All students are expected to respect the privacy of each other by not sharing or using such information outside the course.

Accommodation. Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with the Disability Resource Office at Holmes Hall, Suite 304, Campus Box 7509, 919-515-7653. For more information on NC State's policy on working with students with disabilities, please see the Academic

Accommodations for Students with Disabilities Regulation (REG02.20.01) (<https://policies.ncsu.edu/regulation/reg-02-20-01/>).

Non-Discrimination Policy. NC State provides equal opportunity and affirmative action efforts, and prohibits all forms of unlawful discrimination, harassment, and retaliation ("Prohibited Conduct") that are based upon a person's race, color, religion, sex (including pregnancy), national origin, age (40 or older), disability, gender identity, genetic information, sexual orientation, or veteran status (individually and collectively, "Protected Status"). Additional information as to each Protected Status is included in NCSU REG 04.25.02 (Discrimination, Harassment and Retaliation Complaint Procedure). NC State's policies and regulations covering discrimination, harassment, and retaliation may be accessed at <http://policies.ncsu.edu/policy/pol-04-25-05> or <https://oied.ncsu.edu/divweb/>. Any person who feels that he or she has been the subject of prohibited discrimination, harassment, or retaliation should contact the Office for Equal Opportunity (OEO) at 919-515-3148.

COVID-related information

Due to the COVID-19 pandemic, public health measures continue to be implemented across campus. Students should stay current with these practices and expectations through the [Protect the Pack](https://www.ncsu.edu/coronavirus/) website (<https://www.ncsu.edu/coronavirus/>). The sections below provide expectations and conduct related to COVID-19 issues.

Health and Participation in Class

We are most concerned about your health and the health of your classmates and instructors/TAs.

- If you test positive for COVID-19, or are told by a healthcare provider that you are presumed positive for the virus, you should not attend any face-to-face (F2F) classes or face-to-face component of a hybrid class. Work with your instructor on any adjustments necessary; also follow other university guidelines, including self reporting ([Coronavirus Self Reporting](#)): Self-reporting is not only to help provide support to you, but also to assist in contact tracing for containing the spread of the virus.
- If you feel unwell, even if you have not been knowingly exposed to COVID-19, please do not come to a F2F class or activity.
- If you are in quarantine, have been notified that you may have been exposed to COVID-19, or have a personal or family situation related to COVID-19 that prevents you from attending this course in person (or synchronously), please connect with your instructor to make alternative plans, as necessary.
- If you need to make a request for an academic consideration related to COVID-19, such as a discussion about possible options for remote learning, please talk with your instructor.

Health and Well-Being Resources

These are difficult times, and academic and personal stress are natural results. Everyone is encouraged to [take care of themselves](#) and their peers. If you need additional support, there are many resources on campus to help you:

- Counseling Center ([NCSU Counseling Center](#))
- Student Health Services ([Health Services | Student](#))
- If the personal behavior of a classmate concerns or worries you, either for the classmate's well-being or yours, we encourage you to report this behavior to the NC State CARES team: ([Share a Concern](#)).
- If you or someone you know are experiencing food, housing or financial insecurity, please see the Pack Essentials Program ([Pack Essentials](#)).

Community Standards related to COVID-19

We are all responsible for protecting ourselves and our community. Please see the [community standards](#) and Rule 04.21.01 regarding Personal Safety Requirements Related to COVID-19 [RUL 04.21.01 – Personal Safety Requirements Related to COVID-19 – Policies, Regulations & Rules](#)

Course Expectations Related to COVID-19:

- **Face Coverings:** All members of the NC State academic community are expected to follow all university policies and guidelines, including the [Personal Safety Rule](#) and [community standards](#), for the use of face coverings.
- **Course Attendance:** NC State attendance policies can be found at: [REG 02.20.03 – Attendance Regulations – Policies, Regulations & Rules](#). Please refer to the course's attendance, absence, and deadline policies for additional details. If you are quarantined or otherwise need to miss class because you have been advised that you may have been exposed to COVID-19, you should not be penalized regarding attendance or class participation. However, you will be expected to develop a plan to keep up with your coursework during any such absences. If you become ill with COVID-19, you should follow the steps outlined in the health and participation section above.
- **Technology Requirements:** This course may require particular technologies to complete coursework. Be sure to review the syllabus for these expectations, and see the [syllabus technical requirements](#) for your course. If you need access to additional technological support, please contact the Libraries' Technology Lending Service: ([Technology Lending](#)).

Course Delivery Changes Related to COVID-19

Please be aware that the situation regarding COVID-19 is frequently changing, and the delivery mode of this course could change accordingly, including from in-person to remote. Regardless of the delivery method, we will strive to provide a high-quality learning experience.

NO LONGER AVAILABLE - Grading/Scheduling Changing Options Related to COVID-19

Two policies, enhanced S/U Grading Option and Late Drop, put in place at the beginning of the COVID-19 pandemic have been discontinued.

In some cases, an option may be to request an “incomplete” in the course. If you are experiencing difficult or extenuating circumstances, you should discuss possible options with your instructor and your academic advisor.

Need Help?

If you find yourself in a place where you need help, academically or otherwise, please review these [Step-by-Step Help Topics](#). In addition, the following College of Engineering individuals are very available to assist students with any matter, please reach out to them:

- Dr. Laura Bottomley, Women and Minority Engineering Program, laurab@ncsu.edu
- Ms. Angelitha Daniel, Women and Minority Engineering Program, aldaniel@ncsu.edu
- Dr. David Parish, COE Assistant Dean, dwparrish@ncsu.edu
- Dr. Jerome Lavelle, COE Associate Dean, jplavell@ncsu.edu

Other Important Resources

- **Keep Learning:** [Keep Learning](#)
- **Protect the Pack FAQs:** [Frequently Asked Questions](#) | [Protect the Pack](#)
- **NC State Protect the Pack Resources for Students:** [Resources for Students](#) | [Protect the Pack](#)

- **Academic Success Center** (tutoring, drop in advising, career and wellness advising): [Academic Success Center](#).
- **NC State Keep Learning**, tips for students opting to take courses remotely: [Keep Learning Tips for Remote Learning](#)
- **Introduction to Zoom for students:** <https://youtu.be/5LbPzzPbYEw>
- **Learning with Moodle, a student's guide to using Moodle:** <https://moodle-projects.wolfware.ncsu.edu/course/view.php?id=226>
- **NC State Libraries** [Technology Lending Program](#)

DISCLAIMER Information in this syllabus is subject to minor changes and the possession of it does not ensure that you have all the correct information required to successfully complete the semester. Announcements of changes will be made via e-mail from the class moodle site and students are responsible for reading these e-mails.